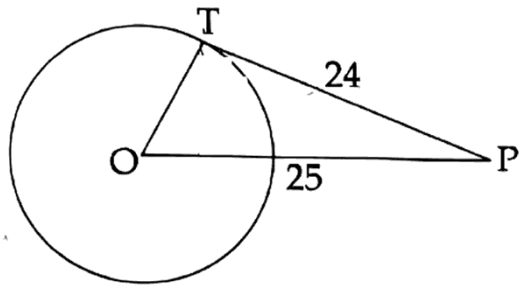
TANGENTS

PYQ • 2021 - 2026

1. 2026 **1 MARK**

Consider the figure and read the following statements.



Statement 1 : The length of the tangent drawn from the point P to the circle is 24 centimetres. If OP is 25 centimetres, then the radius of the circle is 7 centimetres.

Statement 2 : A tangent to a circle is perpendicular to the radius through the point of contact.

Now choose the correct answer from those given below.

(a) Statement 1 is true, statement 2 is false

(b) Statement 1 is false, statement 2 is true

(c) Both statements are true, statement 2 is the reason of statement 1

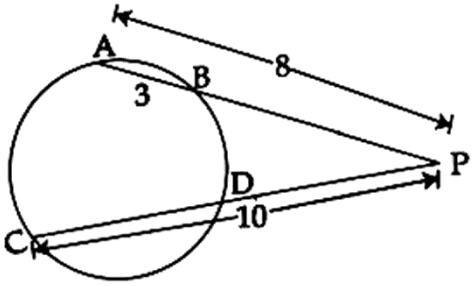
(d) Both statements are true, statement 2 is not the reason of statement 1

2. 2026 **5 MARKS**

Draw a circle of radius 2.5 centimetres. Draw a triangle of two angles 70° and 50° with its sides touching the circle.

3. 2025 **3 MARKS**

In the figure PA = 8 centimetres, AB = 3 centimetres, PC = 10 centimetres.



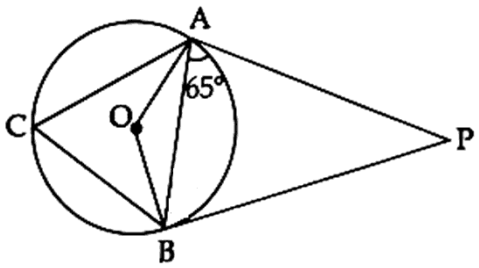
(a) Find the length of PB.

(b) Find the length of PD.

4. 2025 **3 MARKS**

Draw a circle of radius 3 centimetres. Mark a point 9 centimetres away from the centre. Draw tangents from this point to the circle.

5. 2025 **4 MARKS**



In the figure, O is the centre of the circle. PA and PB are tangents, ∠PAB = 65°. Find the measures of the angles below :

(a) ∠PBA = _____

(b) ∠AOB = _____

(c) ∠P = _____

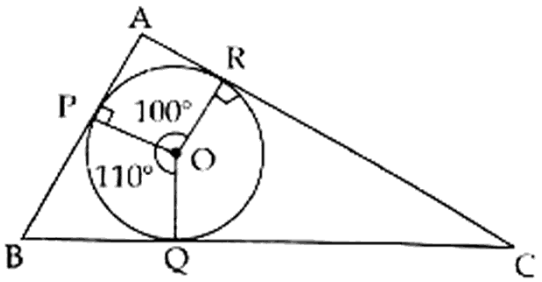
(d) ∠ACB = _____

6. 2025 **5 MARKS**

Draw a circle of radius 2 centimetres. Draw a triangle with two of the angles 55° and 75° with all its sides as the tangents of the circle.

7. 2024 **3 MARKS**

AB, BC and CA are tangents to the circle centred at O, touching the circle at P, Q and R respectively, as shown in the figure.



(a) Find ∠QOR.

(b) Find the angles of triangle ABC.

8. 2024 **4 MARKS**

Draw a circle of radius 2.5 centimetres and mark a point 6 centimetres away from the centre of the circle.

(a) How many tangents can be drawn to the circle from this point ?

(b) Draw the tangents to the circle from this point.

9. 2024 **5 MARKS**

Draw a triangle of sides 5 centimetres, 6 centimetres and 7 centimetres. Draw the incircle of the triangle. Measure the radius of the incircle.

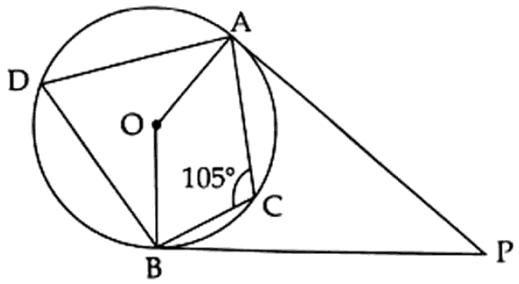
10. 2023 **4 MARKS**

Draw a circle of radius 2.5 centimetres. Mark a point 6.5 centimetres away from the centre.

Draw the tangents to the circle from this point.

Measure and write the lengths of the tangents.

11. 2023 **5 MARKS**



PA and PB are two tangents to the circle centred at O. ∠ACB = 105°. Find the angles given below.

(a) ∠ADB = _____

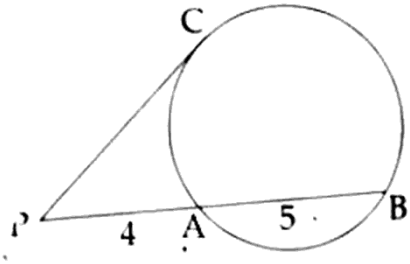
(b) ∠AOB = _____

(c) ∠APB = _____

(d) ∠ABP = _____

(e) ∠ABO = _____

12. 2022 **2 MARKS**

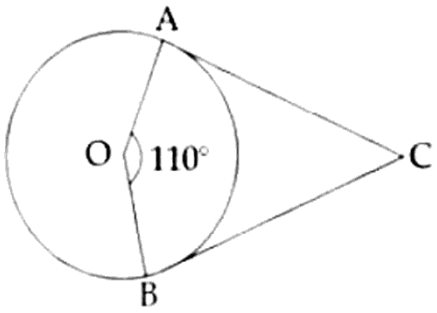


In the figure PA = 4 centimetres, AB = 5 centimetres and PC is a tangent to the circle. Find the length of PC.

13. 2022 **4 MARKS**

Draw a circle of radius 2.5 centimetres and mark a point 6 centimetres away from the centre of the circle. Draw tangents to the circle from this point.

14. 2022 **8 MARKS**



(a) The two tangents AC and BC of the circle with centre O meets at C. What is the measure of ∠OAC ? If ∠AOB = 110° find the measure of ∠ACB. **(2)**

(b) Draw a circle of radius 2.5 centimetres. Draw a triangle with angles 50°, 60°, 70° and all its sides are tangents to this circle. **(6)**

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15. 2021 **3 MARKS**

P is at a distance of 13 centimetres from the centre of a circle of radius 5 centimetres.

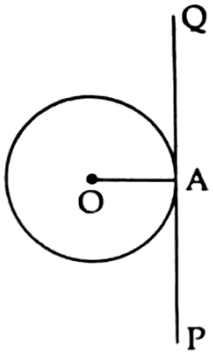
(a) How many tangents can be drawn from the point P to the circle ? **(1)**

(b) Find the lengths of the tangents. **(2)**

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16. 2021 **3 MARKS**

(a) In the figure OA is the radius of the circle. PQ is the tangent through A. What is the measure of ∠OAP ? **(1)**



(b) Draw a circle of radius 3 centimetres and mark a point A on it. Draw the tangent through A. **(2)**

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17. 2021 **4 MARKS**

Draw a circle of radius 3 centimetres. Mark a point P outside the circle at a distance 7 centimetres from the centre. Draw tangents from P to the circle. Measure the length of the tangents.
